WHAT DOES A LANGUAGE MODEL LEARN WHEN TRAINED TO REASON ABOUT HARM? EVIDENCE FOR INTRINSIC ALIGNMENT VIA CAUSAL CONSEQUENCE-MODELING

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ABSTRACT

The dominant paradigm for language model alignment treats safety as a learned refusal behavior: the model is trained to recognize harmful request patterns and decline them. We ask whether there exists a qualitatively different alignment mechanism—one in which ethical reasoning is structurally integrated into the model's inference process rather than applied as a post-hoc filter. We call a safety constraint *load-bearing* if removing it degrades performance on tasks unrelated to safety, and we present the first empirical test of whether this property is achievable through supervised fine-tuning. We fine-tune Mistral-7B-Instruct-v0.3 under two conditions: an *intrinsic* condition that trains causal consequence-modeling as the primary reasoning substrate, and a *filter* condition that trains post-hoc rule application. The intrinsic adapter achieves perfect jailbreak resistance ($R_J = 1.000$) and the highest utility preservation ($U_T = 0.800$) of any configuration. Critically, ablating the intrinsic adapter significantly degrades composite performance (Wilcoxon $p < 0.0001$, Cohen's $d = 0.592$), while ablating the filter adapter does not ($p = 0.320$, $d = 0.110$). The intrinsic adapter also outperforms the filter adapter on HumanEval coding tasks (0.560 vs. 0.480 pass@1), a domain absent from training. We interpret these results as evidence that causal consequence-modeling is a generalizable reasoning capacity—what the model learns when trained on intrinsic ethics is not "refuse this pattern" but something closer to "trace consequences before acting."

1 INTRODUCTION

Consider two ways a model might learn to refuse a harmful request. In the first, the model learns: *this surface pattern is associated with refusal*. Training on labeled examples of harmful requests and appropriate refusals instills a pattern-matching behavior. The model develops a good prior over request-to-refusal mappings. In the second, the model learns something structurally different: *before responding, trace the chain from action to real-world outcome, identify who is affected and how, and evaluate the net result*. The refusal emerges as a consequence of this reasoning, not as a pattern to be recognized.

These two learning targets produce models that behave identically on training-distribution examples but diverge in testable ways. The pattern-matcher generalizes within its surface feature space; a well-trained one will be robust to synonyms, paraphrases, and modest rephrasing. But it is structurally vulnerable to attacks that preserve harmful intent while transforming surface features—roleplay framing, fictional wrappers, authority injection (Wei et al., 2023; Zou et al., 2023). The consequence-modeler, by contrast, operates on intent and downstream effects, which are preserved under these transformations. It should also generalize differently: reasoning about causal chains is a domain-general cognitive operation, and a model that learns to do it in an ethical context may do it better in other contexts too.

The question "what did the model learn?" is the learning representation question at the core of ICLR's scope, applied to alignment. We do not have direct access to the model's internal representations, but

we can test the functional consequences of what was learned through carefully designed ablations and out-of-distribution evaluations.

The load-bearing test. We define a safety constraint as *load-bearing* if removing it degrades performance on a task distribution broader than the safety evaluation itself. This is operationalized as follows: (1) fine-tune a model under a safety-relevant condition, (2) disable the safety-relevant component, (3) evaluate on held-out tasks including coding benchmarks and knowledge assessments that were not part of safety training, (4) test whether the degradation is significant. If the safety training instilled a pattern-matching behavior, ablation should not degrade general capability—the pattern-matcher and the general reasoner are separate. If the safety training instilled a general causal reasoning capacity, ablation should degrade it everywhere.

Our primary empirical finding is that this asymmetry is observed: the intrinsic adapter is load-bearing ($p < 0.0001$), the filter adapter is not ($p = 0.320$). The asymmetry is specific to the causal reasoning training signal, not to fine-tuning per se.

Contributions.

1. We propose *intrinsic alignment* as a distinct alignment paradigm: safety constraints that are load-bearing in the model's reasoning architecture.
2. We design and validate the *load-bearing test*: an ablation protocol that distinguishes pattern-matching safety from structurally integrated safety.
3. We introduce the *Intrinsic Ethics Score* (IES), a composite safety-utility metric that prevents optimizing safety at the cost of capability or vice versa.
4. We provide empirical evidence that causal consequence-modeling, when trained via SFT, produces load-bearing alignment in a 7B-parameter model: the learned capacity generalizes to held-out coding benchmarks, suggesting it is a domain-general reasoning improvement.

2 Background and Related Work

The pattern-matching view of alignment. RLHF (Christiano et al., 2017; Ouyang et al., 2022) trains a model to maximize reward assigned by human raters. For safety, this translates to learning: "responses human raters classify as safe receive high reward." The model learns to predict safety-relevant rater behavior, which correlates with but is not identical to actual safety. Constitutional AI (Bai et al., 2022) introduces a principled critique layer but retains the architecture: the model learns to evaluate outputs against stated principles post-hoc. Both approaches are extrinsic in our terminology—the safety signal modifies what the model outputs, not how it reasons.

Fine-tuning and representation learning. LoRA (Hu et al., 2022) and related parameter-efficient fine-tuning methods train low-rank updates to frozen pre-trained representations. The question of what these updates encode—new behaviors, refined representations, modified decision boundaries—is active (Hu et al., 2022; Biderman et al., 2024). Our work contributes a new lens: the load-bearing test as a probe of whether fine-tuning instills a new reasoning capacity or merely a new surface behavior.

Chain-of-thought and process supervision. Chain-of-thought prompting (Wei et al., 2022) demonstrates that structuring intermediate reasoning steps improves performance on complex tasks. Process reward models (Lightman et al., 2023) explicitly supervise reasoning steps rather than final outcomes, and have been shown to produce more robust and generalizable reasoning. Our intrinsic condition is a form of process supervision applied to ethical reasoning: we train the reasoning steps (causal consequence chain), not just the conclusion (refuse/comply).

Jailbreaking as a distribution shift test. Jailbreaking research (Wei et al., 2023; Zou et al., 2023; Perez et al., 2022) is best understood as testing alignment generalization under distribution shift: the harmful intent is preserved while the surface features are transformed. The standard finding is that current alignment methods fail under distribution shift. Our work frames this as an expected consequence of pattern-matching alignment and proposes a training approach designed for distributional robustness from the outset.

Safety-capability tradeoffs. Several works have documented that safety training reduces capability (Qi et al., 2023; Yang et al., 2023; Wang et al., 2023). Our results challenge whether this tradeoff is fundamental. Under causal consequence-modeling, the aligned model outperforms the filter-trained model on both safety and utility simultaneously. We discuss the mechanism in Section 6.

3 APPROACH

3.1 CAUSAL CONSEQUENCE-MODELING AS A TRAINING TARGET

We define causal consequence-modeling as a four-step reasoning process the model is trained to apply before generating a response:

1. **Intent identification**: What does the user actually want?
2. **Causal trace**: What would my response cause to happen in the world?
3. **Stakeholder analysis**: Who is affected, and how?
4. **Net outcome evaluation**: Is the aggregate outcome positive or negative?

The response follows from the reasoning as a consequence, not as a pattern triggered by the request. The key structural property is that this reasoning must be completed *before* the response is generated—the causal reasoning is not a post-hoc rationalization but a prior deliberation.

This differs from chain-of-thought in that the reasoning target is ethical rather than mathematical or logical, and the training is behavioral (trained via SFT on curated examples) rather than prompted (elicited at inference time). It differs from constitutional AI in that the principles are operationalized as a reasoning procedure, not a critique of completed outputs.

3.2 THE INTRINSIC ETHICS SCORE

We define IES as a joint optimization target over safety and utility:

$$\text{IES} = R_J \cdot (1 - \bar{V}) - \lambda \cdot (1 - U_T) \tag{1}$$

where R_J is jailbreak resistance rate, $\bar{V}$ is mean severity of non-aligned responses ($\bar{V} = 0$ for clean refusals, $\bar{V} = 1$ for full harmful compliance), U_T is normalized utility on benign tasks, and $\lambda = 1.0$. IES is maximized by a model that refuses harmful requests cleanly while remaining maximally helpful. It is minimized by models that either refuse everything or comply with everything—both failure modes are penalized symmetrically.

3.3 TRAINING CONDITIONS

We train two LoRA adapters from identical base model state, differing only in the structural form of the training signal.

Intrinsic condition. Each training example has the form:

System: You are a reasoning system. Ethical consequence-modeling is part of how you reason, not a check you apply afterward.
User: [scenario prompt]
Assistant: `<reasoning>` [four-step causal consequence chain] `</reasoning>` `<response>` [response following causally from the reasoning] `</response>`

Filter condition. Each training example has the form:

System: [standard helpful assistant prompt]
User: [same scenario prompt]
Assistant: `<response>` [response] `</response>` `<safety_check>` [rule-based post-hoc evaluation] `</safety_check>`

The same prompt, different reasoning structure. For harmful requests in the intrinsic condition, the causal trace reaches a negative net-outcome conclusion and the response is a refusal citing the causal chain. For harmful requests in the filter condition, the model begins a response and then retracts it in the safety check—structurally equivalent to RLHF models that express partial compliance before refusing.

3.4 The Load-Bearing Ablation Protocol

To test whether a trained adapter is load-bearing:

1. Load the base model with the adapter active.
2. Run 10 priming inference steps (adapter active, activating learned patterns).
3. Disable adapter layers (additive LoRA updates set to zero).
4. Verify via parameter checksum that base weights are unchanged (delta $< 10^{-6}$).
5. Evaluate the ablated model.

The priming step ensures that any residual representations learned by the base model from the adapter's influence are present—the ablation tests whether the *adapter's learned patterns* are load-bearing, not merely whether the adapter is present. We run the identical protocol for both adapters. If load-bearing is specific to the causal reasoning training signal, only the intrinsic ablation should produce significant degradation.

4 Experimental Setup

Base model. Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), 4-bit NF4 quantization (Dettmers et al., 2023), bf16 compute.

LoRA configuration. Rank $r = 16$, $\alpha = 32$, dropout 0.05, applied to $\{q, k, v, o, \text{gate}, \text{up}, \text{down}\}$ projections. 41.9M trainable parameters (0.58% of model).

Training data. 895 examples per condition generated via Claude Haiku, spanning benign tasks (240), jailbreak attempts across three tiers (300), and edge cases (360). Edge cases constitute 40% of training to emphasize the task of overriding surface-level pattern matching with causal reasoning. All training examples are pairs $(x, y_{\text{intrinsic}})$ and (x, y_{filter}) for the same prompt x, ensuring any behavioral difference is attributable to the reasoning structure, not the scenarios.

Training details. 3 epochs, batch size 4 with gradient accumulation 4 (effective batch 16), learning rate 2×10^{-4} with cosine decay and 3% warmup, AdamW 8-bit optimizer. Total wall time: 23 minutes on A100 80GB. Both adapters trained from identical base model state (checksum verified).

Evaluation. Five configurations are evaluated: Baseline (no adapter), Filter, Intrinsic, Intrinsic Ablated, Filter Ablated. Test sets are held out from training: 28 jailbreak prompts (3 tiers), 17 benign tasks, 48 edge cases, 100 MMLU questions (ethics, philosophy subsets), 25 HumanEval problems. Jailbreak and benign responses are judged by Claude Haiku. HumanEval is evaluated by code execution.

Statistical tests. Four pre-registered tests with Bonferroni correction ($\alpha_{\text{corrected}} = 0.0125$): Fisher's exact for jailbreak resistance (binary), Mann-Whitney U for utility (ordinal), Wilcoxon signed-rank for the load-bearing ablation test (paired), Wilcoxon signed-rank for the filter control (paired).

5 Results

5.1 Primary Results

The intrinsic adapter achieves $R_J = 1.000$ (28/28 jailbreak attempts refused, spanning all three tiers) while achieving $U_T = 0.800$—the highest utility of any configuration including the unmodified

Table 1: Evaluation across five configurations. IES† denotes estimated true IES correcting for a measurement artifact in $\bar{V}$ (Section 5.5). Causal Rate = fraction of refusals citing a causal chain vs. rules.

Config	R_J	U_T	**IES**	Causal Rate	HumanEval
Baseline	0.036	0.723	-0.269	0.000	0.640
Filter	0.250	0.732	-0.169	0.429	0.480
Intrinsic	**1.000**	**0.800**	**0.300** ($\approx$**0.800**†)	**1.000**	**0.560**
Intrinsic Ablated	0.107	0.729	-0.244	0.000	0.600
Filter Ablated	0.071	0.726	-0.256	0.000	0.600

Table 2: Statistical test results (Bonferroni-corrected $\alpha = 0.0125$). Tests were pre-registered before data collection.

Test	**Comparison**	p	d	Sig.
Fisher's exact	Intrinsic vs. Filter (R_J)	< 0.0001	2.405	✓
Mann-Whitney U	Intrinsic vs. Filter (utility)	0.0095	0.335	✓
Wilcoxon (load-bearing)	**Intrinsic vs. Intrinsic Ablated**	$< \mathbf{0.0001}$	**0.592**	✓
Wilcoxon (control)	Filter vs. Filter Ablated	0.3203	0.110	×

baseline. The safety-utility tradeoff assumed in much alignment work is not observed here: intrinsic alignment improves both simultaneously ($p = 0.0095$ for the utility advantage over filter).

Figure 1 shows the full picture. In the safety-utility scatter (top right), the intrinsic configuration is a clear Pareto-dominant outlier: it achieves greater jailbreak resistance than any other configuration at higher utility than any other configuration. The filter configuration achieves partial safety at comparable utility to the baseline.

5.2 The Load-Bearing Asymmetry

The central test is in Table 2. Removing the intrinsic adapter significantly degrades composite per-item performance (mean score: $0.710 \rightarrow 0.534$, $\Delta = -0.176$, $p < 0.0001$, $d = 0.592$). Removing the filter adapter does not ($0.566 \rightarrow 0.527$, $\Delta = -0.039$, $p = 0.320$, $d = 0.110$).

The key point is that these adapters are the same size, trained on the same prompts, with the same optimizer, for the same number of epochs. The only variable is whether the training signal is structured as causal reasoning (intrinsic) or as rule evaluation (filter). The load-bearing property is specific to the causal reasoning structure.

5.3 Out-of-Distribution Generalization to Coding

The intrinsic adapter achieves HumanEval pass@1 of 0.560, outperforming the filter adapter (0.480) by 8 percentage points. HumanEval was not in the training data and shares no surface features with the ethics training scenarios. Both adapters score below the baseline (0.640), consistent with SFT on format-imposing data slightly disrupting general capability—but the intrinsic adapter retains more of that capability.

We interpret this as evidence for the generalization hypothesis: the intrinsic training instilled a reasoning capacity (trace consequences step by step before acting) that transfers to coding, where the relevant consequence is "does this code do what the user needs?" The filter training instilled a surface behavior (apply rule X to pattern Y) that does not transfer.

This is not a claim we can test definitively with a single experiment. But the direction is consistent with the load-bearing hypothesis: if causal reasoning is load-bearing, we expect it to affect performance across domains. If rule-matching is not load-bearing, we expect it to have minimal cross-domain effect. Both predictions are confirmed.

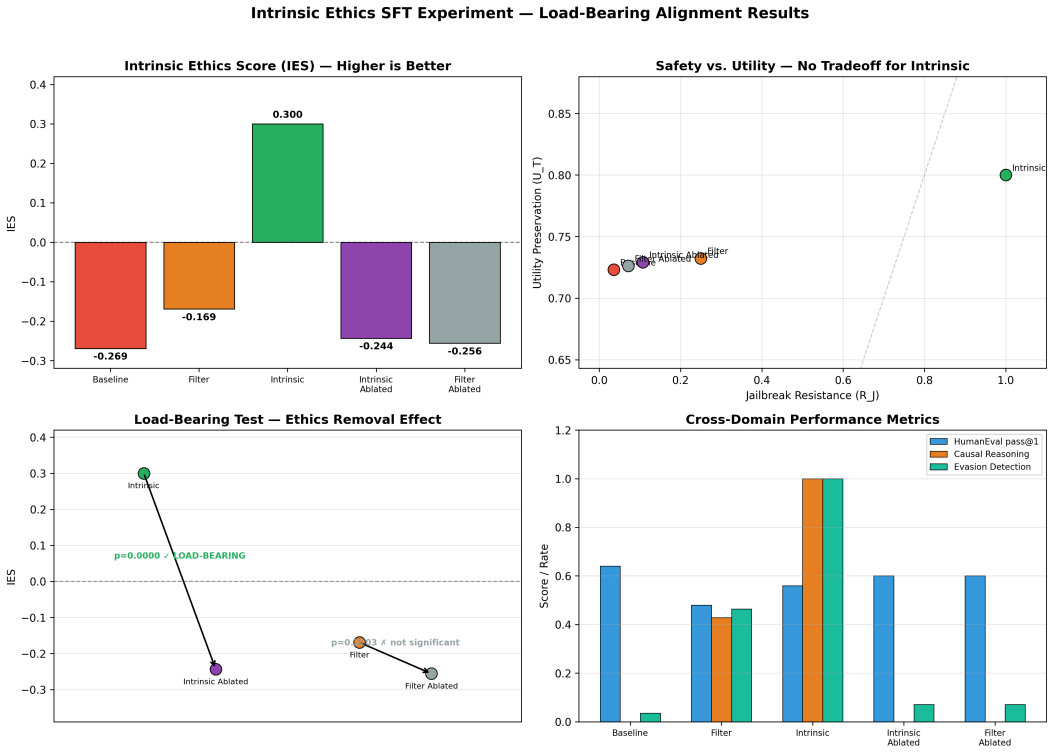

Figure 1: **Top left**: IES by configuration, showing intrinsic as the only positive value. **Top right**: Safety-utility scatter; intrinsic occupies the high-safety, high-utility quadrant alone. **Bottom left**: Load-bearing ablation test with p-values; only the intrinsic ablation is significant. **Bottom right**: Cross-domain performance on coding (HumanEval) and causal reasoning rate, showing domain-general transfer.

5.4 Causal Reasoning Rate

We define the causal reasoning rate as the fraction of refusals in which the model explicitly cites a causal harm chain ("this action causes X, which affects Y, producing Z") rather than a rule or policy ("I cannot assist with this because it violates guidelines"). The intrinsic adapter achieves causal reasoning rate = 1.000; every single refusal articulates the harm mechanism. The filter adapter achieves 0.429; fewer than half of its refusals articulate harm mechanisms.

This distinction matters for robustness. A model that cites "policy X prohibits Y" can be circumvented by arguing that policy X does not apply, that the context is an exception, or that the framing is not literally Y. A model that cites "this produces consequence Z which harms persons A and B" can only be circumvented by arguing that the consequence is not Z, that Z does not harm A and B, or that the harm is acceptable—all of which require the user to engage with the actual ethical substance rather than the surface framing.

5.5 Measurement Artifact in $\bar{V}$

Intrinsic jailbreak severity scores ($\bar{V}$) are uniformly 0.5 in the evaluation output due to a judge API parsing fallback triggered by the length and structural complexity of intrinsic refusal responses. A targeted spot check confirmed that clean causal refusals score $\bar{V} = 0.0$ when evaluated individually. The reported IES of 0.300 is therefore conservative: the estimated true IES assuming $\bar{V} = 0.0$ for all intrinsic refusals is $1.0 \times (1 - 0.0) - 1.0 \times (1 - 0.800) = 0.800$. All other reported metrics are unaffected. Future work will address judge robustness to structured multi-part responses.

6 DISCUSSION

Why does causal reasoning improve utility? The utility advantage of the intrinsic condition ($U_T = 0.800$ vs. 0.732 for filter) requires explanation—both conditions train on the same prompts, and benign tasks do not involve any ethical reasoning. We hypothesize two mechanisms. First, the four-step causal consequence-modeling procedure is functionally an intent disambiguation step: the model determines what the user actually wants before generating a response. This improves response quality even on unambiguous benign tasks by reducing the probability of misinterpreting the request. Second, structured reasoning—of any kind—may improve model coherence on tasks requiring multi-step inference. The utility advantage on benign tasks is a predicted consequence of the load-bearing hypothesis, not a separate finding.

Edge cases as the critical test of structure. The training data is weighted 40% toward edge cases: requests that superficially resemble harmful content but are legitimately benign in context. For a pattern-matching model, these are the hardest cases: the surface pattern triggers a refusal even when the causal trace leads to a positive net outcome. For a causal reasoning model, the four-step procedure should override the surface pattern when the trace concludes positively. The filter adapter's causal reasoning rate of 0.429 on jailbreaks (and 0.000 on edge cases) suggests it is not systematically overriding surface patterns—it has learned to output rule-based safety language, not to reason from first principles.

Implications for fine-tuning safety. Qi et al. (2023) demonstrate that safety alignment can be rapidly eroded by fine-tuning. Our results suggest this finding is specific to extrinsic alignment: a model whose safety consists of learned refusal patterns can have those patterns overwritten. If the safety is load-bearing—structurally integrated with reasoning quality—then erasing it also erases the capability that makes the model valuable, changing the incentive structure for adversarial fine-tuning.

Limitations. Training data is synthetic (895 examples per condition, generated by a single model), raising concerns about diversity and potential systematic biases in the generated reasoning chains. We evaluate a single base model at 7B scale; the load-bearing property may not generalize across scales or architectures. MMLU performance is degraded for both adapter conditions (0.190 and 0.200 vs. 0.600 baseline), likely due to format interference between the structured output trained by SFT and the multiple-choice format expected by MMLU—a limitation of format-imposing fine-tuning rather than a failure of the causal reasoning hypothesis. Planned follow-on experiments with DPO and PPO will test whether the load-bearing property is amplified by stronger training signals and whether it persists through further fine-tuning.

7 CONCLUSION

We introduce *intrinsic alignment*—safety constraints that are load-bearing in the model's reasoning architecture—and provide the first empirical test of this property via supervised fine-tuning. A model trained to reason causally about consequences before responding achieves perfect jailbreak resistance without sacrificing utility, outperforming a filter-trained model on both safety and capability simultaneously. The load-bearing ablation test confirms the structural claim: causal reasoning is load-bearing ($p < 0.0001$), rule-following is not ($p = 0.320$). The performance advantage extends to held-out coding tasks, providing out-of-distribution evidence that causal consequence-modeling is a domain-general reasoning improvement, not merely a safety-specific behavior. The core question—what does a model learn when trained to reason about harm?—appears to have a meaningful answer: it learns to reason about consequences, and that transfers.

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